Linchen Zhang

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Education

University of California, San Diego San Diego, CA
Ph.D. Candidate in Economics *Sep 2021 – Jun 2027 (Expected)*

Cornell University Ithaca, NY
B.A. in Computer Science, Economics (Magna Cum Laude), and Mathematics *Aug 2017 – May 2021*

Fields of Interest

Microeconomic Theory, Behavioral and Experimental Economics

Working Papers

Recognizing Dominance through Local Improvement *(Job Market Paper)*

- *Abstract:* Participants in strategy-proof mechanisms often fail to play their dominant strategies. Most existing simplicity concepts attribute these failures to the cost of contingent reasoning and lower the cost by comparing the worst case of the dominant strategy with the best case of an alternative. I argue that a participant can instead compare a strategy with its neighbors, such as the bids one step above and below, and move to a better neighbor until none remains. I formalize this idea with a neighbor relation on each strategy set and call a mechanism local improvement solvable if every such sequence of moves ends at a dominant strategy. Strategy-proofness is the criterion under the complete neighbor relation. Under their natural neighbor relations, the criterion holds in the sealed-bid second-price auction, the multi-unit Vickrey auction, and, in rich environments, serial dictatorship, top trading cycles, and deferred acceptance. In extensive-form mechanisms without duplicate dominant strategies, the criterion under one-situation neighbors is strict action-proofness, which yields a one-shot deviation principle for dominance. This criterion and obvious strategy-proofness classify the pick-a-number game, chess, and Sudoku in opposite ways. Among local improvement solvable mechanisms that implement the same outcomes, the model predicts more dominant-strategy play when the local comparison is salient.

Market Order Mechanisms (with Ben Brooks and Songzi Du)

- If the market order mechanisms satisfy vanishing price impact and vanishing span, they can guarantee competitive outcomes in the large, regardless of information structures and equilibria.

Coarse Revenue Guarantee in First Price Auction and Beyond

- I characterize the worst-case revenue in the first-price auction over all Bayes coarse correlated equilibria.

Work in Progress

Conditional Dominance and Rationalizability with Information-Based Continuation Reasoning (with Joel Watson)

- Draft coming soon!

Strategic Obliviousness (with Emanuel Vespa and Andreas Ziegler)

- Third wave of experiments completed; Analyzing data and preparing for a draft.

Professional / Academic Activities

Academic Presentations

2026: Econometric Society Asian-China Meeting, Southwest Economics Theory Conference, Asian School in Theory, UCLA Student Theory Conference, UCSD TBE Workshop

2025: 36th Stony Brook Game Theory Conference, Caltech Student Theory Conference, UCSD TBE Workshop

2024: EC'24 (poster), 35th Stony Brook Game Theory Conference, GAIMSS'24, UCSD TBE Workshop

Discussions and Service

2026: Program Committee, ACM Conference on Economics and Computation (EC)

2026: Panelist, Virtual Seminars in Economic Theory, on Lea Nagel's talk "Designing Simpler Mechanisms: Transparency and As-If Dominance"

Awards / Fellowships

Clive Granger Research Fellowship, UC San Diego
awarded for the most promising graduate student research — *2025*

Summer Research Fellowship, UC San Diego — *2022, 2023*

Regents Fellowship, UC San Diego — *2021*

Honorable Mention, Mathematical Contest in Modeling (MCM) — *2019*

Top 500, William Lowell Putnam Mathematical Competition — *2018*

Omicron Delta Epsilon (International Economics Honor Society), Cornell University

Dean's List (All semesters), Cornell University — *2017 – 2021*

Teaching Experience

Instructor of Record at UC San Diego

ECON 5: Data Analysis in Social Science — *Fall 2024, Fall 2025*

Teaching Assistant at UC San Diego

ECON 1: Principles of Microeconomics — *Fall 2022, Fall 2023*

ECON 4: Financial Accounting — *Winter 2023, Spring 2024*

ECON 5: Data Analysis in Social Science	*Spring 2025*
ECON 100B: Microeconomics B	*Winter 2024, Winter 2026*
ECON 149: Market Design	*Winter 2024, Winter 2025*
ECON 173B: Corporate Finance	*Spring 2023*

SKILLS / OTHER INFO

- **Programming:** Python, STATA, MATLAB, R, LaTeX
- **Languages:** Mandarin Chinese (native), English (proficient)

REFERENCES

Professor Songzi Du (Chair)
Department of Economics, UC San Diego
sodu@ucsd.edu

Professor Aram Grigoryan
Department of Economics, UC San Diego
a2grigoryan@ucsd.edu

Professor Joel Watson
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Professor Emanuel Vespa
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